

MALYADIP PAL

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B.Tech (Hons.) in Instrumentation Engineering, IIT Kharagpur

Education

Year	Degree/Exam	Institute	CGPA/Marks
2027	B.Tech (Hons.)	Indian Institute of Technology, Kharagpur	9.19 / 10
2023	Class XII (CBSE)	Infocus India Public School	96%
2021	Class X (WBBSE)	Paschimpara High School	96.85%

Experience

Member of Technical Staff Intern | DevRev

May '26 – Jul '26

Enterprise platform for support, product management, and CRM

- Designed and shipped a **fine-grained object-sharing feature** on core work objects, enabling granular access control and collaborative sharing across the platform
- Built a **gRPC annotation framework** enforcing API-level access control, preventing internal-only endpoints from being exposed externally across multiple microservices
- Developed a **cross-service authorization resolver** mapping entities to their access-control policies, enabling consistent permission checks in a distributed microservice architecture
- Hardened platform security by deprecating legacy authorization flows and migrating services to a centralized, **privilege-based access-control model**

Competitions

National 2nd Runner Up | Legacy Document Processing | Hack The Future '25 [\[GitHub\]](#)

Feb '25 – Mar '25

- Built an optimized pipeline to extract text, tables, and images from scanned government files using **EasyOCR** and **PyMuPDF**
- Engineered a **GeminiVLM**-powered table extractor; integrated **RAG** agent for semantic search and SQL analytics on PostgreSQL
- Developed a unified **React.js** + **Flask** frontend for query-driven exploration and document visualization

Silver Winner | GC Vibemeter | OpenSoft 2025, General Championship [\[GitHub\]](#)

Mar '25 – Apr '25

- Designed an **ANOVA-inspired scoring model** to flag at-risk employees; built **LangGraph** multi-agent chatbot for wellness
- Developed full-stack app with **React** + **FastAPI**, **WebSocket** real-time chat, and automated Slack alerts via cron jobs

Projects

SmartKrishi – AI-Powered Agricultural Assistant [\[GitHub\]](#)

Aug '25 – Sep '25

- Architected multi-service backend with **FastAPI**, **SQLAlchemy**, **PostgreSQL**; streaming AI with chain-of-thought reasoning
- Built **SMS/WhatsApp fallback** for low-network areas; **Firebase OTP auth** for inclusive access; **Telegram** bot integration
- Developed **React** + **TypeScript** frontend with streaming chat, image-based crop disease detection, and Zustand state management

Research Paper Classifier | KDSH Hackathon 2025 | National Finalist [\[GitHub\]](#)

Dec '24 – Jan '25

- Built **voting classifier** (SVM, LR, NB + TF-IDF) achieving **95% accuracy**; **Gemini RAG pipeline** for conference recommendation
- Implemented custom **multithreaded vector store** with real-time Google Drive sync for efficient semantic retrieval

Technical Skills

Languages: C, C++, Python, Go, JavaScript, TypeScript, HTML/CSS, SQL

Frameworks: React, Express.js, FastAPI, Flask, Tailwind CSS, LangChain, LangGraph, NumPy, Pandas, Scikit-learn

Tools & Infra: Git, Node.js, PostgreSQL, MongoDB, MySQL, gRPC, Protocol Buffers, Firebase, Docker, Postman

Achievements

- Expert on Codeforces** (max 1618); Ranked **116/37,000+** in CF Round 1032 | **4-Star CodeChef** (peak 1803)
- Secured **AIR 2967** in JEE Advanced 2023 (top 1.5%); **99.59 percentile** in JEE Mains (top 0.42%)
- IAPT Merit Certificate** for top 1% rank in NSEP 2022 (41,454 participants nationwide)

Relevant Coursework

Systems: Computer Architecture, Operating Systems, Embedded Systems, Digital Electronic Circuits, Signals & Systems

CS/AI: Programming and Data Structures, High Performance Parallel Programming

Math: Linear Algebra, Probability and Statistics, Advanced Calculus

Extra-Curricular

- Member of **RK Hall OpenSoft Team**; represented hall in Inter-Hall competition for collaborative software development
- NSS Student Volunteer**, IIT Kharagpur (2023-24); mental health and community awareness campaigns